

✓ The abstract classes in java can _____ *

1/1

- ☒ Implement constructors ✓
- ☐ Can't implement constructor
- ☐ Can implement only unimplemented methods
- ☐ Can't implement any type of constructor

✓ In Java, declaring a class abstract is useful *

1/1

- ☐ To prevent developers from further extending the class.
- ☒ When it doesn't make sense to have objects of that class. ✓
- ☐ To force developers to extend the class not to use its capabilities.
- ☐ When it makes sense to have objects of that class.

✓ What will be the output of the following Java code? *

1/1

```
class A
{
    int i;
    void display()
    {
        System.out.println(i);
    }
}
class B extends A
{
    int j;
    void display()
    {
        System.out.println(j);
    }
}
class method_overriding
{
    public static void main(String args[])
    {
        B obj = new B();
        obj.i=1;
        obj.j=2;
        obj.display();
    }
}
```

☐ 0

☐ 1

☒ 2



☐ Compilation Error

✓ Which of the following statements regarding abstract classes are true? * 1/1

- ☐ An abstract class can be extended.
- ☐ A subclass of a non-abstract superclass can be abstract.
- ☐ A subclass can override a concrete method in a superclass to declare it abstract.
- ☒ All of the above ✓

✓ Suppose A is an abstract class, B is a concrete subclass of A, and both A and B have a default constructor. Which of the following is correct? *1/1

1. A a = new A();
2. A a = new B();
3. B b = new A();
4. B b = new B();

☐ 1 and 2

☒ 2 and 4 ✓

☐ 3 and 4

☐ 1 and 3

✓ Which of the following differentiates between overloaded functions and overridden functions? *1/1

- ☐ Overloading is a dynamic or runtime binding and overridden is a static or compile time binding.
- ☒ Overloading is a static or compile time binding and overriding is dynamic or runtime binding. ✓
- ☐ Redefining a function in a friend class is called overloading, while redefining a function in a derived class is called as overridden function.
- ☐ Redefining a function in a derived class is called function overloading, while redefining a function in a friend class is called function overriding.

✓ If a class inheriting an abstract class does not define all of its function then it will be known as? *1/1

- ☒ Abstract ✓
- ☐ A simple class
- ☐ Static class
- ☐ None of the mentioned

✓ Which of these is correct about passing an argument by call-by-value process?

*1/1

- ☒ Copy of argument is made into the formal parameter of the subroutine. ✓
- ☐ Reference to original argument is passed to formal parameter of the subroutine.
- ☐ Copy of argument is made into the formal parameter of the subroutine and changes made on parameters of subroutine have effect on original argument.
- ☐ Reference to original argument is passed to formal parameter of the subroutine and changes made on parameters of subroutine have effect on original argument.

✓ What is the process of defining a method in terms of itself, that is a method that calls itself?

*1/1

- ☐ Polymorphism
- ☐ Abstraction
- ☐ Encapsulation
- ☒ Recursion ✓

✓ What is the output of the below Java program with multiple methods? *

1/1

```
public class MethodOverloading1
{
    void show(int a, char b)
    {
        System.out.println("KING KONG");
    }

    void show(char a, int b)
    {
        System.out.println("JIM JAM");
    }

    public static void main(String[] args)
    {
        MethodOverloading1 m = new MethodOverloading1();
        m.show(10, 'A');
        m.show('B', 10);
    }
}
```

- ☐ JIM JAM JIM JAM
- ☐ KING KONG KING KONG
- ☒ KING KONG JIM JAM
- ☐ compiler error



✓ To successfully overload a method in Java, the return types must be ____.* 1/1

- ☐ Same
- ☐ Different
- ☒ Same but using superclass or subclass types also work
- ☐ None



✓ Which of these is supported by method overriding in Java? *

1/1

- ☐ Abstraction
- ☐ Encapsulation
- ☒ Polymorphism
- ☐ None of the mentioned



✓ Which of these can be overloaded? *

1/1

- ☐ Methods
- ☐ Constructors
- ☒ All of the mentioned
- ☐ None of the mentioned



✓ What is the process of defining two or more methods within same class that have same name but different parameters declaration? *1/1

- ☒ method overloading ✓
- ☐ method overriding
- ☐ method hiding
- ☐ none of the mentioned

✓ What is the process of defining a method in a subclass having same name & type signature as a method in its superclass? *1/1

- ☐ Method overloading
- ☒ Method overriding ✓
- ☐ Method hiding
- ☐ None of the mentioned